MILITARY SPECIFICATION

RING, SUPPORT ASSEMBLY, MP CARTRIDGE

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers the requirements and quality assurance provisions for support rings molded from high density polyethylene for packaging XM830 and XM831 120mm (MP) cartridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. - Unless otherwise specified, the following specifications, standards, and handbooks of the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

THIS DOCUMENT CONTAINS PROPRIETARY DATA AND WAS FURNISHED TO THE UNITED STATES GOVERNMENT PURSUANT TO CONTRACT NUMBER DAAK10-79-C-0094 BETWEEN RHEINMETALL GmbH AND THE DEPARTMENT OF THE ARMY DATED 22 FEB. 1979 AND THE RECEIVER IS SUBJECT TO THE PROVISIONS THEREOF AND AGREES TO THE OBLIGATIONS OF SUCH CONTRACT.

BENEFICIAL COMMENTS (RECOMMENDATIONS, ADDITIONS, DELETIONS) AND ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT SHOULD BE ADDRESSED TO: COMMANDER, US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND, ATTN. DRDAR-QA, DOVER, NEW JERSEY 07801 BY USING THE SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARING AT THE END OF THIS DOCUMENT OR BY LETTER.

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-A-2550	Ammunition, General Specification For
MIL-A-48078	Ammunition, Standard Quality Assurance Provision, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings and publications. - The following Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

12527200	Ring, Support Assembly (MP Cartridge)
12527201	Ring, Support (Halved) Assembly (MP Cartridge)

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12527202	Ring, Support (MP Cartridge)
12527203	Clip, Tension

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-1238	Measuring Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM-D-1505	Density of Plastics by the Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

2.3 Order of precedence. - In the event of a conflict between the text of this specification and the referenced specifications cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Components. - The components shall comply with all requirements specified on applicable drawings and specifications referenced herein.

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3.3 Physical properties. - The support rings shall not crack, split, warp or break when exposed to a temperature of 71°C for 100 hours when tested as specified in 4.5.3.

3.4 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, surface defects, flash, air bubbles, blisters and dirt. The cleaning method shall not be injurious to the parts, nor shall the parts be contaminated by the cleaning agent.

3.5 First article inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 5 complete sets of parts from each die or mold used to produce the molded part and be so identified as to permit tracing each item to the die or mold in which it was molded.

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

DOD-R-64046 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Support Ring Assembly, MP Cartridge				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Tension Clip (12527203)				
	Examination for defects	10			4.4.2.1
	Support Ring, MP Cartridge (12527202)				
	Examination for defects	10			4.4.2.2
	Support Ring (Halved) Assembly, MP Cartridge (12527201)				
	Examination for defects	10			4.4.2.3
	Support Ring Assembly, MP Cartridge (12527200)				
	Examination for defects	10			4.4.2.4
	Melt index	10			4.5.1
	Density	10			4.5.2
	Thermal stress cracking resistance	10			4.5.3
	Color	10			4.5.4

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4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, lots shall contain:

a. Material from not more than one lot from one manufacturer.
b. Material of one color designation.

4.4.2 Examination. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (see MIL-A-48078, paragraph 4.4.2).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

DOD-R-64046 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Clip, Tension				12527203
					NEXT HIGHER ASSEMBLY 12527201
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Verify wire material	Ea. lot		3.1	Vendor cert.
102	Diameter of wire		0.40%	3.2	Gage
103	Protective finish incorrect, missing or incomplete		0.40%	3.2	Visual
104	Contour between B and C incorrect		0.40%	3.2	Gage
Minor					
201	Color incorrect		0.65%	3.2	Visual
202	Contour between A and B incorrect		0.65%	3.2	Gage
203	Contour between C and D incorrect		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

DOD-R-64046 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12527202
4.4.2.2	Ring, Support (MP Cartridge)				NEXT HIGHER ASSEMBLY 12527201
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Verify material	Ea. lot		3.1	Vendor cert.
102	Overall length (or height)	1/		3.2	Gage
103	Cracks, splits or voids		0.40%	3.2	Visual
104	Thickness of radial center support ring	1/		3.2	Gage
105	Thickness of radial support cross rib	1/		3.2	Gage
Minor					
201	Sprue above flush within I.D. or beyond O.D. of ring		0.65%	3.2	Visual/Manual
202	Color incorrect		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

1/ Two per shift

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

DOD-R-64046 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.3	Ring, Support (Halved) Assembly (MP Cartridge)		1 OF 1		12527201
					NEXT HIGHER ASSEMBLY 12527200
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components damaged or missing		0.40%	3.2	Visual
102	Components improperly assembled		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

DOD-R-64046 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12527200
4.4.2.4	Ring, Support Assembly, MP Cartridge				NEXT HIGHER ASSEMBLY 12527260
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembled outside diameter 1/		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

1/ This inspection conducted after assembly on 120mm mandrel

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4.4.3 Testing. - Tests shall be in accordance with the Quality Conformance Inspection sheets.

4.4.3.1 Cure time. - Determination of compliance with dimension and tolerance requirements shall be delayed for a minimum of 48 hours after molding.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections prescribed in this specification is identified in the examination (4.4.2) and Test Methods (4.5) paragraphs herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Melt index. - Melt index shall be determined in accordance with ASTM-D-1238.

4.5.2 Density. - The density of the test specimen shall be determined in accordance with ASTM-D-1505.

4.5.3 Thermal stress cracking resistance. - Subject the support ring to 71 $\pm$ 1°C for 100 hours. Immediately following the 100 hour time period, visually examine each support ring while warm, for evidence of cracks, splits, warping or other deformations.

4.5.4 Color. - Examine the specimen, on all sides, using daylight illumination to determine compliance with 3.2.

5. PACKAGING

5.1 Cleaning. The following items shall be cleaned with process C-1 of MIL-P-116, if required.

Support Ring Assembly, MP Cartridge - 12527200

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5.2 Drying. - The items listed in paragraph 5.1 shall be dried in accordance with MIL-P-116, if required.

5.3 Preservation. - Not required.

5.4 Unit and intermediate packaging

5.4.1 Level A. - All items listed in paragraph 5.1 shall be unit and intermediate packaged in accordance with Level A of MIL-STD-1169.

5.4.2 Level B. - All items listed in paragraph 5.1 shall be unit and intermediate packaged in accordance with Level B of MIL-STD-1169.

5.4.3 Commercial. - All items listed shall be packaged in accordance with MIL-STD-1188.

5.5 Packing

5.5.1 Level A. - All packages in paragraph 5.4.1 shall be packed in accordance with Level A of MIL-STD-1169.

5.5.2 Level B. - All packages in paragraph 5.4.2 shall be packed in accordance with Level B of MIL-STD-1169.

5.5.3 Commercial. - All packages in paragraph 5.4.3 shall be packed in accordance with MIL-STD-1188.

5.6 Marking

5.6.1 Level A. - Marking to be in accordance with MIL-STD-1169.

5.6.2 Level B. - Marking to be in accordance with MIL-STD-1169.

5.6.3 Commercial. - Marking to be in accordance with MIL-STD-1188.

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5.7 Shipping. - When assemblies from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots, clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. - The supports covered by this specification are intended for use in supporting XM830 and XM831 projectiles in ammunition containers.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval (see MIL-A-48078). - Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801.

6.4 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-A, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5 Molding instructions. - The particular material selected as to density and melt index should be injection molded so as to meet the requirements of this specification. It must be noted that selection of a material complying with the density and melt index requirements does not necessarily insure compliance with the performance requirements of this specification nor relieve the molder of his responsibility for molding in such a fashion that the support will meet the requirement.

6.6 Density specimen. - Saw speed in excess of 500 to 600 lineal feet per minute have been found to be unsatisfactory as excess speed can melt the material and greatly change the density.

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6.7 Temperatures for continuous use or storage of the molded supports should not exceed 176 degrees F.

6.8 Definition of packaging levels

6.8.1 Level A. - Full Military Protection for unrestricted (OCONUS) shipment and storage.

6.8.2 Level B. - Limited Military Protection for limited storage and (OCONUS) shipment.

6.8.3 Level C. - For interplant (CONUS) shipment and very limited storage.

6.8.4 For more detailed interpretation of packaging and packing levels, consult AR-700-15.

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